

DATA STRUCTURES & ANALYSIS OF ALGORITHMS (Theory & Lab)

Subject Codes: BMC205 & BMC254

PART 1: THEORY SYLLABUS

Subject Code: BMC205

Subject Name: Data Structures & Analysis of Algorithms

Periods: 4-0-0

Course Outcomes (CO)

At the end of the course, the student will be able to:

CO	Description	Bloom's Level
CO 1	Explain the concept of data structure, abstract data types, algorithms, analysis of algorithms and basic data organization schemes such as arrays and linked lists.	K2
CO 2	Describe the applications of stacks and queues and implement various operations on them using arrays and linked lists.	K3
CO 3	Describe the properties of graphs and trees and implement various operations such as searching and traversal on them.	K3
CO 4	Compare incremental and divide-and-conquer approaches of designing algorithms for problems such as sorting and searching.	K4
CO 5	Apply and analyze various design approaches such as Divide-and-Conquer, greedy and dynamic for problem solving.	K4

Detailed Syllabus

Unit I (08 Lectures)

Introduction to Data Structure: Data, Entity, Information, Difference between Data and Information, Data type, Built-in data type, Abstract data type, Definition of data structures, Types of Data Structures: Linear and Non-Linear Data Structure.

Introduction to Algorithms: Definition of Algorithms, Difference between algorithm and programs, properties of algorithm, Algorithm Design Techniques, Performance Analysis of Algorithms, Complexity of various code structures, Order of Growth, Asymptotic Notations.

Arrays: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D, 2-D Array, Application of arrays, Sparse Matrices and their representations.

Linked Lists: Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List: Insertion, Deletion, Traversal, Polynomial Representation and Addition, Subtraction & Multiplications of Single variable.

Unit II (08 Lectures)

Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Iteration and Recursion- Principles of recursion, Tail recursion, Removal of recursion, Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers.

Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.

Searching: Concept of Searching, Sequential search, Index Sequential Search, Binary Search. Concept of Hashing & Collision resolution Techniques used in Hashing.

Unit III (08 Lectures)

Sorting: Insertion Sort, Selection Sort, Bubble Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time: Counting Sort and Bucket Sort.

Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth First Search and Breadth First Search, Connected Component.

Unit IV (08 Lectures)

Trees: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Complete Binary Tree, Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search Tree. Threaded Binary trees, Huffman coding using Binary Tree, AVL Tree and B Tree.

Unit V (08 Lectures)

Divide and Conquer: With Examples Such as Merge Sort, Quick Sort, Matrix Multiplication: Strassen's Algorithm.

Greedy Programming: Dijkstra Algorithm, Prims and Kruskal algorithm.

Dynamic Programming: Bellman Ford Algorithm, All-pair Shortest Path: Warshal Algorithm, Longest Common Sub-sequence.

Suggested Readings

1. Cormen T. H., Leiserson C. E., Rivest R. L. and Stein C., "Introduction to Algorithms", PHI.

2. Horowitz E., Sahni S. and Rajasekharan S., "Fundamentals of Computer Algorithms", Universities Press.
3. Dave P. H. and Dave H. B, "Design and Analysis of Algorithms", Pearson Education.
4. Lipschuts S., "Theory and Problems of Data Structures (Schaum's Series)", Tata McGraw-Hill.
5. Goyal K. K., Sharma S. and Gupta A., "Data Structures and Analysis of Algorithms", HP Hamilton.
6. Samanta D., "Classic Data Structures", Prentice Hall India.
7. Goodrich M. T. and Tomassia R., "Algorithm Design: Foundations, Analysis and Internet Examples", John Wiley and sons.
8. Sridhar S., "Design and Analysis of Algorithms", Oxford Univ. Press.
9. Aho A., Ullman J. and Hopcroft J., "Design and Analysis of algorithms", Pearson Education.1920
- 10.Neapolitan R. and K. Naimipour, "Foundations of Algorithms", Jones and Bartlett Student Edition. 212223

2: LAB SYLLABUS

Subject Code: BMC254

Subject Name: Data Structures & Analysis of Algorithms Lab

Course Outcomes (CO)

At the end of the course, the student will be able to:

CO	Description	Bloom's Level
CO 1	Write and execute programs to implement various searching and sorting algorithms.	K3
CO 2	Write and execute programs to implement various operations on two-dimensional arrays.	K3
CO 3	Implement various operations of Stacks and Queues using both arrays and linked lists data structures.	K3
CO 4	Implement graph algorithm to solve the problem of minimum spanning tree.	K3

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List of Experiments (Program in C or C++)

1. To implement addition and multiplication of two 2D arrays.
2. To transpose a 2D array.
3. To implement stack using array.
4. To implement queue using array.
5. To implement circular queue using array.

6. To implement stack using linked list.
7. To implement queue using linked list.
8. To implement BFS using linked list.
9. To implement DFS using linked list.
- 10.To implement Linear Search.
- 11.To implement Binary Search.
- 12.To implement Bubble Sorting.
- 13.To implement Selection Sorting.
- 14.To implement Insertion Sorting.
- 15.To implement Merge Sorting.
- 16.To implement Heap Sorting.
- 17.To implement Matrix Multiplication by Strassen's algorithm.
- 18.Find Minimum Spanning Tree using Kruskal's Algorithm.

Note: The instructor may add/delete/modify experiments, wherever he/she feels in a justified manner.